

RECENT ADVANCES ON THE EQUILIBRIUM FLOW PROBLEM

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SIAM Conference on Optimization (OP26)
MS137 – Optimization over Probability Space
University of Edinburgh – June 3, 2026

This talk draws on:

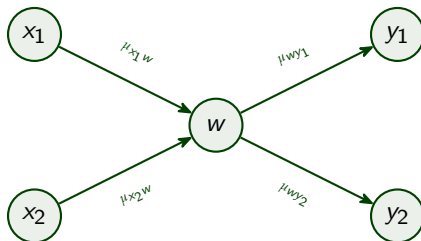
- [GSV1]** A. Galichon, L. Samuelson, L. Vernet, *Unified gross substitutes and inverse isotonicity for equilibrium problems*, Theoretical Economics **20** (2025), 941–971. arXiv:2207.06731
- [GSV2]** A. Galichon, L. Samuelson, L. Vernet, *The existence of equilibrium flows*, Journal of Mechanism and Institution Design **9**(1) (2024), 55–81. arXiv:2209.04426
- [GJ]** A. Galichon, A. Jacquet, *Substitutability, equilibrium transport, and matching models*, 2024. arXiv:2405.07628

1. THE EQUILIBRIUM FLOW PROBLEM
2. THE REGULARIZED PROBLEM
3. BACK TO THE UNREGULARIZED PROBLEM

THE EQUILIBRIUM FLOW PROBLEM

A directed network (Z, A) : nodes $z \in Z$, arcs $(x, y) \in A$.

- ▶ $\mu_{xy} \geq 0$: flow of the commodity on arc xy .
- ▶ p_z : price of the commodity at node z .



Source nodes, destination nodes, and intermediate nodes (products, locations, ...).

- ▶ $G_{xy}(p_y)$: the *breakeven purchase price* at x if the resale price at y is p_y .
- ▶ *Example.* If the transportation cost on arc xy is c_{xy} , then

$$G_{xy}(p_y) = p_y - c_{xy}$$

– a higher cost lowers the breakeven price.

- ▶ Consider the carry trade: buy at x , resell at y . Its profit is

$$\pi_{xy}(p) = G_{xy}(p_y) - p_x,$$

decreasing in the purchase price p_x .

READING THE CONDITIONS ON π_{xy}

- ▶ $\pi_{xy}(p) > 0$: a strictly profitable arbitrage on arc xy .
- ▶ $\pi_{xy}(p) = 0$: the carry trade just breaks even.
- ▶ $\pi_{xy}(p) < 0$: the carry trade loses money.

- ▶ q_z : net outflow at node z ; $q = (q_z)_{z \in Z}$.
- ▶ ∇ : the discrete gradient = node–arc incidence matrix, with

$$(\nabla p)_{xy} = p_y - p_x.$$

- ▶ Conservation of mass at every node:

$$\nabla^\top \mu = q.$$

∇ maps node prices $p \in \mathbb{R}^Z$ to arc differences in $\mathbb{R}^A$; its transpose $\nabla^\top$ maps arc flows $\mu \in \mathbb{R}^A$ to net outflows in $\mathbb{R}^Z$.

An *equilibrium flow* is a pair (μ, p) with $\mu \geq 0$ such that:

EQUILIBRIUM CONDITIONS

PRIMAL FEASIBILITY conservation of mass:

$$\nabla^\top \mu = q.$$

DUAL FEASIBILITY no strictly positive profit:

$$G_{xy}(p_y) - p_x \leq 0 \quad \forall (x, y) \in A.$$

COMPLEMENTARITY no mass on negative-profit arcs:

$$\mu_{xy} > 0 \implies G_{xy}(p_y) - p_x = 0.$$

Prices p play the role of (dual) potentials; flows μ are supported on the breakeven arcs.

The equilibrium flow problem nests several classical models:

- ▶ **Optimal transport** – bipartite, additive $G_{xy}(p_y) = p_y - c_{xy}$.
- ▶ **Minimum-cost flow problem** – additive $G_{xy}(p_y) = p_y - c_{xy}$ on a general network.
- ▶ **Stable matchings** – bipartite network with transfers (Demange–Gale) or without (Gale–Shapley).
- ▶ **Scheduling problem** – arcs encoding precedence/timing constraints.

REMARK

The equilibrium flow problem is *not* an optimization problem in general. It reduces to one (a linear program: min-cost flow / optimal transport) precisely when G is **additive**, $G_{xy}(p_y) = p_y - c_{xy}$.

THE REGULARIZED PROBLEM

Replace the complementarity system by a smooth flow:

$$\mu_{xy} = M_{xy}(p) = \exp(G_{xy}(p_y) - p_x).$$

The equilibrium condition becomes

$$\nabla^\top M(p) = q.$$

M-FUNCTION STRUCTURE

Once a *ground node* is excluded (prices normalized there), the reduced map $p \mapsto \nabla^\top M(p)$ is an *M-function*: off-diagonally antitone (a *Z-function*) and nonreversing. (On the full space the constant-shift direction makes it degenerate.) This is the structure we leverage for existence, uniqueness, and computation.

How do we solve $\nabla^\top M(p) = q$? Sweep through the nodes, clearing one at a time.

GAUSS-SEIDEL (COORDINATE SWEEP)

At node z , solve the scalar balance equation for p_z , holding the other prices fixed,

$$(\nabla^\top M(p))_z = q_z,$$

then pass to the next node, *immediately reusing the updated prices*.

- ▶ M -functions are *exactly* the class of maps that make Gauss-Seidel work: the sweep converges *monotonically* to the solution from any sub- or supersolution.
- ▶ (Jacobi updates every node at once from the old prices; Gauss-Seidel uses the freshest values.) Classical: Rheinboldt (1970); Ortega & Rheinboldt (1970).

Economic reading of the coordinate sweep:

- ▶ Each node adjusts its own price to *clear its own balance* (net flow = q_z), taking the other prices as given.
- ▶ This is a discrete **tâtonnement**, a price-adjustment process market by market.

WHY SUBSTITUTES MAKE IT CONVERGE

The off-diagonal antitonicity of an M -function is the gross-substitutes property: raising the price at one node does not raise net supply elsewhere. So when a node raises its price to clear, the others' imbalances move in one consistent direction — the sweep neither cycles nor overshoots.

This is the discrete analogue of the classical convergence of Walrasian tâtonnement under gross substitutes.

EXISTENCE OF EQUILIBRIUM PRICES [GSV2]

Under an **acyclicity** condition on the network, the regularized system $\nabla^\top M(p) = q$ admits a solution p .

- ▶ Proof generalizes *Hall's marriage theorem* to the flow setting.
- ▶ Feasibility of q is governed by a profitability / Hall-type condition on cuts of the network.

UNIQUENESS [GJ]

Under a **connectedness** condition, the solution p of $\nabla^\top M(p) = q$ is unique *up to a constant*.

- ▶ Only price *differences* $p_x - p_y$ enter G_{xy} , hence the constant.
- ▶ Fix it by **normalizing** the price at a chosen *ground node* z_0 , e.g. $p_{z_0} = 0$.
- ▶ Uniqueness then follows from the inverse-isotonicity theory of M -functions.

COMPARATIVE STATICS

With p normalized at the ground node z_0 , the equilibrium price p is *isotone* in the net-outflow vector q (restricted to $z \neq z_0$):

$$q \leq q' \implies p \leq p'.$$

- ▶ Direct consequence of inverse isotonicity of the M -function $\nabla^\top M(\cdot)$.
- ▶ Increasing demand at a node (weakly) raises equilibrium prices everywhere.

- ▶ Solve $\nabla^\top M(p) = q$ by **Gauss–Seidel**: sweep the nodes, updating each p_z to clear node z and reusing fresh values.
- ▶ For the EFP the node- z update is a single monotone scalar equation in p_z (closed-form in the linear/OT case), so each sweep is cheap.
- ▶ By the M -function structure, the sweep converges from any sub-/supersolution [GJ].

LINK WITH SINKHORN

For the bipartite / optimal-transport special case, the coordinate updates are exactly the **Sinkhorn** iterations on the regularized transport problem.

BACK TO THE UNREGULARIZED PROBLEM

Drop the exponential regularization and return to the complementarity problem.

In this talk we focus on what becomes of two properties:

- ▶ **Uniqueness** of the equilibrium price.
- ▶ **Isotonicity** of the price in the net outflow.

Neither survives verbatim – the equilibrium price set is now genuinely set-valued.

THE TWO PROPERTIES, GENERALIZED

- ▶ **Uniqueness** $\rightsquigarrow$ the set of equilibrium prices is a *sublattice* of the price space.
- ▶ **Isotonicity** $\rightsquigarrow$ monotonicity in *Veinott's strong set order*.

Recall, for sets S, S' , the strong set order $S \leq S'$:

$$s \in S, s' \in S' \implies s \wedge s' \in S, s \vee s' \in S'.$$

This is the right notion of “increasing” for set-valued maps.

Fix a dual-feasible price p . Define the *supply correspondence*

$$Q(p) = \{ \nabla^\top \mu : \mu \geq 0, \mu_{xy} > 0 \Rightarrow G_{xy}(p_y) - p_x = 0 \},$$

the net outflows q supported on the breakeven arcs at p .

UNIFIED GROSS SUBSTITUTES (UGS) – [GSV1]

$Q : P \rightrightarrows \mathbb{R}^Z$ satisfies UGS if for all p, p' and all $q \in Q(p)$, $q' \in Q(p')$, there exist $q^\wedge \in Q(p \wedge p')$ and $q^\vee \in Q(p \vee p')$ such that, for every z ,

$$\begin{aligned} p_z \leq p'_z &\implies q_z \leq q_z^\wedge \text{ and } q_z^\vee \leq q'_z, \\ p'_z < p_z &\implies q'_z \leq q_z^\wedge \text{ and } q_z^\vee \leq q_z. \end{aligned}$$

NONREVERSING CORRESPONDENCE – [GSV1]

$Q : P \rightrightarrows \mathbb{R}^Z$ is *nonreversing* if, for all p, p' and all $q \in Q(p)$, $q' \in Q(p')$,

$$q \leq q' \text{ and } p \geq p' \implies q \in Q(p') \text{ and } q' \in Q(p).$$

- ▶ Rules out a full reversal of order: a higher price giving a lower quantity is possible only if the two points are interchangeable.
- ▶ The milder of the two conditions. Here it holds *automatically*: every $q \in Q(p)$ satisfies $\sum_{z \in Z} q_z = 0$ (constant aggregate output $\implies$ nonreversing).

Q IS AN $M0$ -CORRESPONDENCE – [GSV1]

The equilibrium supply correspondence Q satisfies *unified gross substitutes* and *nonreversingness*; equivalently, Q is an **$M0$ -correspondence**. Consequently, for every q :

- ▶ the equilibrium price set $Q^{-1}(q)$ is a *sublattice* of P ;
- ▶ $Q^{-1}(q)$ is *increasing in q* in Veinott's strong set order.

Inverse isotonicity of M -matrices, Berry–Gandhi–Haile connected substitutes, and Topkis' theorem all appear as special cases.

- ▶ The **equilibrium flow problem**: a complementarity problem on a network (Z, A) unifying min-cost flow, assignment, matching, and hedonic models.
- ▶ **Regularized** ($M_{xy} = \exp(G_{xy}(p_y) - p_x)$): existence under acyclicity [GSV2], uniqueness up to normalization under connectedness [GJ], isotone comparative statics, Gauss–Seidel/Sinkhorn computation.
- ▶ **Unregularized**: uniqueness $\rightsquigarrow$ sublattice structure; isotonicity $\rightsquigarrow$ Veinott order; the supply correspondence Q is an $M0$ -correspondence [GSV1].

THANK YOU.